

# Script-Controlled ACL – Code Layout, Readability and Reusability

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| Date         | 30 September 2026                                                   |
| Team ID      | SWTID-2026-6203                                                     |
| Project Name | Script-Controlled ACL – Restrict Record Access Based on Field Value |

## Script-Controlled ACL – Code Layout, Readability and Reusability:

This document evaluates the ServiceNow ACL implementation in terms of structure, readability and reusability. The project uses role-based access, Branch-based record restriction, separate CRUD ACLs and script-controlled access decisions.

### Code Layout Checklist:

| S.No | Code Quality Parameter   | Description                                                      | Followed (Yes / No / Partial) | Remarks                                      |
|------|--------------------------|------------------------------------------------------------------|-------------------------------|----------------------------------------------|
| 1    | Consistent Configuration | ACL settings and role assignments are maintained consistently.   | Yes                           | Roles are clearly assigned.                  |
| 2    | Proper Table Structure   | Institution Details fields are logically organized.              | Yes                           | Fields follow project requirements.          |
| 3    | Meaningful Names         | Table, fields and ACL names clearly reflect their purpose.       | Yes                           | Names are clear.                             |
| 4    | ACL / Script Names       | ACL records and scripts are descriptively configured.            | Yes                           | CRUD ACLs are separated.                     |
| 5    | Code Clarity             | Access script clearly represents allow/deny logic.               | Yes                           | Admin and role checks are understandable.    |
| 6    | Modular Design           | Access control is divided into separate ACL operations.          | Yes                           | READ, CREATE, WRITE and DELETE are separate. |
| 7    | No Redundant Logic       | Unnecessary duplicate access-control logic is avoided.           | Yes                           | Each ACL has a specific purpose.             |
| 8    | Access Handling          | Unauthorized users are denied when conditions are not satisfied. | Yes                           | Role and Branch checks control access.       |

### Reusable Components / Modules:

| S.No | Component / Module Name     | Language / Technology              | Where Reused                  | Reusability Level (High / Medium / Low) |
|------|-----------------------------|------------------------------------|-------------------------------|-----------------------------------------|
| 1    | Read ACL + Branch Condition | ServiceNow ACL / Script            | Record-level view restriction | High                                    |
| 2    | Create ACL                  | ServiceNow ACL                     | Record creation control       | High                                    |
| 3    | Write ACL                   | ServiceNow ACL                     | Record update control         | High                                    |
| 4    | Delete ACL                  | ServiceNow ACL                     | Record deletion control       | High                                    |
| 5    | Role-Based Access Script    | ServiceNow Scripting / GlideSystem | Role and administrator checks | High                                    |

# Script-Controlled ACL – Code Layout, Readability and Reusability

## Overall Code Quality Assessment:

| Aspect                   | Rating (1-5) | Comments                                                                                     |
|--------------------------|--------------|----------------------------------------------------------------------------------------------|
| Code Layout & Structure  | 5            | ACL operations are separated clearly into Read, Create, Write and Delete controls.           |
| Readability              | 5            | Role names, field names and access logic are easy to understand.                             |
| Reusability              | 5            | The ACL and role-based approach can be reused for other tables and field-based restrictions. |
| Documentation / Comments | 4            | Project steps and access conditions are documented clearly.                                  |
| Overall Score            | 5            | The implementation is structured, readable and reusable for record-level security.           |

## Project-Specific Summary:

The Script-Controlled ACL project uses ServiceNow ACLs to control record access based on user roles and the Branch field. bb1 controls Read access for EEE records, bb2 controls Create, bb3 controls Write/Edit, and bb4 controls Delete, while administrators retain full access. This separation makes the access-control design easy to maintain and reuse.